

IAL Chemistry Unit 6 (WCH16/01) — Concise A* Study Note

Practical Skills in Chemistry II · built from 20 past papers and mark schemes, June 2019 → June 2026

The examiners' **ACCEPT / IGNORE / REJECT** rulings are kept wherever they decide a mark. June and October 2019 are the legacy WCH06 papers, used where the content is the same.

Contents

§0 The paper at a glance

Part A — Inorganic tests and transition metals (§1–§6)

Part B — Organic tests and spectra (§7–§9)

Part C — Titrations and other quantitative work (§10–§16)

Part D — Kinetics practicals (§17–§19)

Part E — Organic preparation and purification (§20–§27)

Part F — Hazards and safety (§28)

Part G — Exam technique (§29–§33)

0. THE PAPER AT A GLANCE

Time / marks	1 h 20 min, 50 marks , 3–5 questions; bring a ruler
Almost every paper	inorganic ion / transition-metal identification · organic functional-group tests (often + NMR / MS / IR) · a titration calculation · an organic preparation (reflux → separate → dry → distil or recrystallise → melting point → % yield)
Most papers	a kinetics practical with a graph (clock reaction, sampling + titration, half-lives, Arrhenius)
Sometimes	enthalpy of solution, electrochemical cell, combustion analysis, thermal decomposition, water of crystallisation

The six rules of observation answers

1. Say **"precipitate"** for every solid that forms — with its **colour**. Never call a solution a precipitate ("orange solution", not "orange ppt").
2. Give **both the before and after** colour for a change: "orange → green".

3. **Use the scoring colour words:** "(pale) **pink**" for the KMnO_4 end point (purple rejected); "**dark / deep blue**" for copper–ammonia (just "blue" rejected); "**colourless**" (never "clear"); "**pale yellow ppt**" for iodoform (not "bright yellow").
 4. **Fumes and smoke:** HCl is **misty / steamy fumes**; $\text{NH}_3 + \text{HCl}$ gives **white smoke**; NO_2 is **brown fumes** (not "brown smoke"). Don't swap them.
 5. **Negative results:** "**no change / stays blue / stays orange**" — "none" or "no result" is penalised.
 6. **The list principle:** an extra **wrong** observation cancels a right one. Write only what you are sure of.
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PART A — INORGANIC TESTS AND TRANSITION METALS

1. Gas tests

Gas	Test	Result
NH_3	damp red litmus held in the vapour (not dipped in)	turns blue ; or white smoke with conc HCl on a glass rod
CO_2	limewater	cloudy / milky
O_2	glowing splint	relights
H_2	lighted splint	squeaky pop
HCl	damp blue litmus / NH_3	red / white smoke (NH_4Cl)
water vapour	cobalt chloride paper	blue → pink (reject red); condensation at the mouth of the tube

Name the **gas** as NH_3 and the **ion** as NH_4^+ — the schemes reject swapping them ("ammonia ion" rejected).

2. Anion tests

Ion	Test	Positive result
CO_3^{2-} / HCO_3^-	add dilute acid	effervescence ; gas turns limewater milky
SO_4^{2-}	dilute HCl (or HNO_3) then $\text{BaCl}_2(\text{aq})$	white precipitate (BaSO_4)
Cl^-	dilute HNO_3 then $\text{AgNO}_3(\text{aq})$	white ppt, dissolves in dilute NH_3
Br^-	same	cream ppt, insoluble in dilute NH_3, dissolves in conc NH_3
I^-	same	yellow ppt, insoluble in conc NH_3
NO_3^-	NaOH + aluminium , warm	NH_3 (red litmus → blue)

Ion	Test	Positive result
CH_3COO^-	warm with conc H_2SO_4 , then add ethanol	vinegar smell → fruity smell (ester)

Why acidify first: the acid **removes other ions that would also precipitate** — CO_3^{2-} , HCO_3^- , SO_3^{2-} (with Ba^{2+}) or OH^- / CO_3^{2-} (with Ag^+).

- **Never acidify a sulfate test with H_2SO_4** — it adds sulfate, so the test is always positive (0 marks). Never use HCl before AgNO_3 .
- AgCl / AgBr "**colours hard to distinguish**" (not "the same") → confirm with **dilute NH_3** .
- If another precipitate is present (e.g. $\text{Fe}(\text{OH})_2$ with NH_3), **separate the silver halide first**, or the other precipitate obscures the result.
- **Conc H_2SO_4 + a chloride** → **steamy fumes** of HCl (not white smoke).

3. Cation tests

Flame tests: Li^+ red · Na^+ **yellow** · K^+ **lilac** · Ca^{2+} **orange-red (brick red)** · Sr^{2+} red · Ba^{2+} **green** · Cu^{2+} blue-green. Write the **ion**, not the metal.

NH_4^+ : warm with $\text{NaOH}(\text{aq})$ (or CaO) → NH_3 turns damp red litmus blue.

Adding $\text{NaOH}(\text{aq})$ or $\text{NH}_3(\text{aq})$ — a few drops, then excess

Add **dropwise with a teat (dropping) pipette**, then in excess.

Ion (colour)	Few drops NaOH / NH_3	Excess NaOH	Excess NH_3
Cu^{2+} (blue)	pale blue ppt $\text{Cu}(\text{OH})_2$	no change	dissolves → deep / dark blue solution $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$
Fe^{2+} (pale green)	green ppt $\text{Fe}(\text{OH})_2$ — turns brown on standing (oxidised by air to $\text{Fe}(\text{OH})_3$)	no change	no change
Fe^{3+} (yellow-brown)	(red-/orange-)brown ppt $\text{Fe}(\text{OH})_3$	no change	no change
Cr^{3+} (green / violet)	grey-green ppt $\text{Cr}(\text{OH})_3$	dissolves → dark green solution $[\text{Cr}(\text{OH})_6]^{3-}$	dissolves → violet / purple solution $[\text{Cr}(\text{NH}_3)_6]^{3+}$
Mn^{2+} (pale pink)	off-white / pale brown ppt — darkens in air (oxidised)	no change	no change

Cobalt: pink $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ + conc HCl $\rightarrow$ **blue** $[\text{CoCl}_4]^{2-}$.

Iron: Fe^{2+} + oxidant (ClO_3^- , H_2O_2 , air) $\rightarrow$ **green** $\rightarrow$ **yellow / yellow-brown** (reject red). Fe^{3+} oxidises I^- to I_2 (**brown solution**).

Manganese: MnO_4^- **purple**; Mn^{2+} very pale pink / colourless; **MnO_2 brown** (forms if too little acid in KMnO_4 titrations).

Vanadium (reduction by zinc in acid; heat to speed it up):

Oxidation state	Ion	Colour
+5	VO_2^+	yellow
+4	VO^{2+}	blue
+3	V^{3+}	green
+2	V^{2+}	violet / purple (reject lilac)

- The **first green** is a **mixture of yellow VO_2^+ and blue VO^{2+}** , not V^{3+} .
- V^{2+} left in air turns green: **oxidised to V^{3+} by O_2** ; O_2 is not a strong enough oxidant to go further.
- **Deduce an oxidation state** from a KMnO_4 titration by **counting electrons**: $\text{mol e}^- = \text{mol MnO}_4^- \times 5$, then $\div \text{mol V}$.

5. Water of crystallisation and thermal decomposition

Finding x in a hydrate (4 marks): weigh the sample $\rightarrow$ heat in a **crucible / evaporating basin** (not a beaker or test tube) $\rightarrow$ **heat to constant mass** $\rightarrow$ mass of water = **difference**.

$x = (\text{mol water}) \div (\text{mol anhydrous salt})$; or $x = (\text{Mr hydrate} - \text{Mr anhydrous}) \div 18$. Answers: **6** (Mohr's salt), **2** ($(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$), **4** ($\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$), **2** ($\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$).

Heating a hydrated salt — observations: **condensation / droplets** at the mouth of the tube; cobalt chloride paper **blue** $\rightarrow$ **pink**; solid **dissolves in its own water of crystallisation**. (Reject "effervescence".)

Heating a Group 2 / transition-metal nitrate: **brown fumes** (NO_2); O_2 **relights a glowing splint**; nitrates are **oxidising** — a splint in contact with the solid can make it **burn uncontrollably**.

6. Identifying a salt — putting it together

1. Colour of solid / solution $\rightarrow$ likely transition metal.
2. Flame test or NaOH / NH_3 $\rightarrow$ cation.
3. Anion tests (acidify first) $\rightarrow$ anion.

4. Combine as a **neutral formula**: e.g. $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$, $\text{NH}_4\text{Cr}(\text{SO}_4)_2$, $\text{NH}_4\text{Fe}(\text{SO}_4)_2$, BaI_2 , CuCrO_4 , CuCl_2 , $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$.
5. If a **name and a formula** are both given, **both must be correct**; give the **oxidation state** ("iron(II) nitrate", not "iron nitrate").

Ionic equations (with state symbols when asked): $\text{Fe}^{2+} + 2\text{OH}^- \rightarrow \text{Fe}(\text{OH})_2$ · $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$ · $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4$ · $2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$. No uncanceled spectator ions or electrons.

PART B — ORGANIC TESTS AND SPECTRA

7. Functional-group tests

Test	Positive result	Shows
bromine water	orange / brown → colourless (reject "clear")	C=C
acidified KMnO_4	purple → colourless (reject violet)	C=C
burn in air	very smoky (sooty) flame	benzene ring (high C:H)
sodium	effervescence (H_2)	–OH (alcohol or acid)
PCl_5	misty / steamy fumes of HCl (turn blue litmus red)	–OH
Na_2CO_3 / NaHCO_3 (named in full)	effervescence; gas turns limewater milky	–COOH
acidified $\text{K}_2\text{Cr}_2\text{O}_7$, warm	orange → green	1° or 2° alcohol, or aldehyde
2,4-DNPH (Brady's)	yellow / orange / red precipitate (reject brick red, brown)	C=O — aldehyde or ketone
Tollens', warm	silver mirror / grey-black solid (not just "mirror")	aldehyde
Fehling's / Benedict's, warm	blue solution → (brick-)red precipitate (Cu_2O)	aldehyde
I_2 + NaOH, warm (or KI + NaClO)	pale yellow precipitate (CHI_3), antiseptic smell	$\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$
carboxylic acid + conc H_2SO_4 , warm	sweet / fruity smell	alcohol (ester forms)
water, then conc NH_3 on a glass rod	misty fumes, then white smoke	acyl chloride
Universal Indicator	blue / purple	amine (alkaline)
$\text{CuSO}_4(\text{aq})$ + amine	pale blue ppt → deep blue solution	amine

Deductions the schemes reward:

- DNPH positive + dichromate **no change** (or Fehling's stays blue) → **ketone**. + iodoform positive → **methyl ketone $\text{CH}_3\text{CO}-$** .
- Na fizzes but dichromate stays orange → **tertiary alcohol**.
- PCl_5 fumes but **no** NaHCO_3 fizz → **alcohol**, not acid.
- Iodoform positive but **not** a ketone → **$\text{CH}_3\text{CH}(\text{OH})-$** secondary alcohol, or **ethanal**.
- Reject "hydroxide / OH^- " for the hydroxyl group; "benzene / phenyl" for an alkene; "carbonyl" alone when "aldehyde or ketone" is needed.

Detecting an ester in an acid mixture: add NaHCO_3 to **neutralise the acid**, whose smell **masks** the fruity ester smell.

Why phenol reacts with Cl_2 / Br_2 without a catalyst: an **O lone pair** is **delocalised into the ring**, **increasing the ring's electron density**, so it is more susceptible to electrophilic attack.

8. Distinguishing pairs — model answers (reagent + both results)

Pair	Reagent	Results
aldehyde vs ketone (propanal / propanone, pentanal / pentan-3-one)	Tollens' / Fehling's / acidified dichromate, warm	aldehyde: silver mirror / red ppt / orange → green ; ketone: no change
methyl ketone vs other ketone (pentan-2-one / pentan-3-one)	I_2 + NaOH, warm	pale yellow ppt / no change
ethanal vs propanal	I_2 + NaOH	ethanal pale yellow ppt ; propanal no change
acid vs alcohol (propanoic acid / propan-1-ol)	Na_2CO_3 / NaHCO_3 / Mg, or acidified dichromate	acid effervescence ; alcohol orange → green
Fe^{2+} vs Cr^{3+}	NaOH(aq), then excess	both green ppt; $\text{Cr}(\text{OH})_3$ dissolves in excess , $\text{Fe}(\text{OH})_2$ doesn't (goes brown)
valerenic ($\text{C}=\text{C}$) vs valeric acid	bromine water	decolourises / stays orange

The observation mark **depends on the right reagent** (or a near miss). Na and PCl_5 **cannot** separate an acid from an alcohol (both react).

Identifying a carbonyl by its derivative: react with **2,4-DNPH** → **recrystallise** the precipitate → **measure its melting temperature** → compare with data-book values.

9. Spectra in the practical paper

Mass spectrum: M^+ gives M_r ; fragments need a **+** charge: 15 CH_3^+ , 29 C_2H_5^+ / CHO^+ , 43 CH_3CO^+ , 45 COOH^+ , 59 COOCH_3^+ , 77 C_6H_5^+ .

IR (cm^{-1}): alcohol O–H 3750–3200 · **acid O–H 3300–2500 (broad)** · N–H 3500–3300 (amine) / 3500–3140 (amide) — they overlap · **ester C=O 1750–1735** · acid C=O 1725–1700 · **amide C=O 1700–1630**. Identify the **bond and its range**.

^{13}C NMR: number of peaks = carbon environments (**symmetry**); C=O ~160–220 ppm.

^1H NMR: peaks = H environments; **areas** = ratio of H; **$n + 1$ splitting** from H on the **adjacent** carbon; **singlet = no adjacent H**.

Pattern	Means
triplet (3H) + quartet (2H)	CH_3CH_2-
doublet (6H) + septet (1H)	$(\text{CH}_3)_2\text{CH}-$
9H singlet	$(\text{CH}_3)_3\text{C}-$
singlet ~2.1–2.4	$\text{CH}_3\text{CO}-$
quartet ~4.1 / septet ~5.0	CH_2 or CH bonded to ester O
~9–10	$-\text{CHO}$
~10–12	$-\text{COOH}$

Worked identities from the papers: $(\text{CH}_3)_3\text{CCOCH}_3$ (2 singlets 9:3) · hexane-2,5-dione (2 singlets) · $(\text{CH}_3)_2\text{CHCOOH}$ (doublet 6H + septet) · propan-2-yl propanoate · 2,2-dimethylpropanal (9H + 1H) · pentan-3-one (triplet + quartet) · 2-methylpropan-2-ol (2 peaks 9:1) vs butan-2-ol (5 peaks 3:2:1:1:3).

Combustion analysis: mass C = $m(\text{CO}_2) \times 12/44$; mass H = $m(\text{H}_2\text{O}) \times 2/18$; **O by difference** → mole ratio → empirical formula → $M^+ \div$ empirical mass. Water absorbed by **anhydrous CaCl_2** , CO_2 by **soda lime** (not limewater). Answers: $\text{C}_2\text{H}_4\text{O} \rightarrow \text{C}_4\text{H}_8\text{O}_2$; $\text{C}_8\text{H}_8\text{O}$ (phenylethanone).

PART C — TITRATIONS AND OTHER QUANTITATIVE WORK

10. Titration technique

Choosing apparatus

Need	Use	Reject
accurately known volume (aliquot, sample)	volumetric pipette + filler	measuring cylinder, burette for sampling
variable / non-standard volumes (7.5, 11.0 cm ³)	burette or graduated pipette	"just pipette", volumetric pipette
excess reagent / approximate volume (acid, KI)	measuring cylinder — quicker, precision not needed	—
adding dropwise	teat (dropping) pipette	"small amounts"

Making a standard solution (3 marks): dissolve the weighed solid in distilled water in a **beaker** → transfer the solution **and the washings** to a **volumetric flask** → make up **to the mark with distilled water** → **stopper and invert** to mix.

Accurate end point (any 3): add **dropwise near the end point** · **swirl** · **white tile** · rinse the flask walls / burette tip with **distilled water** · read the **bottom of the meniscus at eye level** · **rinse the burette with the titrant** (so it isn't **diluted** by residual water).

Reading the burette from the top of the liquid: the titre is **unchanged**, because it is the **difference between two readings** taken the same way.

11. Indicators and end points

Titration	Indicator / end point
strong acid → alkali (e.g. HCl into Ca(OH) ₂)	methyl orange: yellow → orange (reject yellow → red), or phenolphthalein pink → colourless
weak acid ← NaOH (ethanedioic acid, ethanoic acid)	phenolphthalein: colourless → pink (methyl orange / litmus / UI rejected)
jump in pH 3.5–6.5	bromocresol green or methyl red

Titration	Indicator / end point
KMnO_4	self-indicating: colourless → (first permanent pale) pink (reject purple)
Ce(IV)	self-indicating: colourless → pale yellow
thiosulfate / iodine	starch , added when the solution is pale yellow / straw (not "yellow", not "brown"); end point blue-black → colourless

- Starch is added late because at high $[\text{I}_2]$ it forms an **insoluble complex**, making the end point hard to see.
- **KMnO_4 titrations:** acidify with **dilute H_2SO_4 only** — **HCl** is oxidised to Cl_2 (titre too big); **HNO_3** oxidises Fe^{2+} . Too little acid → **brown MnO_2** . KMnO_4 goes in the **burette**.
- **KMnO_4 + ethanedioate** is warmed ($\sim 60^\circ\text{C}$) because of a **high E_a** ; then **Mn^{2+} produced autocatalyses** it.

12. Mean titre and uncertainty

- Record every burette reading to **2 d.p.** (ending in 0 or 5).
- **Mean titre = mean of concordant titres only** (within $0.10\text{--}0.20\text{ cm}^3$); **exclude the rough and any outlier**. Keep **2 d.p.**: 12.65 (not 12.7); 9.50 (not 9.65, the mean of all three).
- **% uncertainty = (uncertainty × number of readings) ÷ value × 100:**
- burette titre: **$2 \times 0.05 \div \text{titre} \times 100$** (e.g. $0.10 \div 20.10 = 0.50\%$; forgetting $\times 2$ gives 0.25%, rejected);
- temperature change: **$2 \times 0.25 \div \Delta T \times 100$** ($0.5 \div 12.5 = 4\%$);
- pipette: $0.06 \div 10.0 = 0.6\%$ (one reading).
- **Smaller titres** (e.g. using 25.0 cm^3 aliquots of a diluted solution) → **greater % uncertainty**; but aliquots allow **repeat titrations** → **concordant results**.

13. Calculation chains

Always: moles of titrant → **ratio** → moles in the aliquot → **× scale-up** ($\times 10$ for 25.0 cm^3 of 250 cm^3 ; $\times 40$ for 1000 cm^3) → mass / concentration / % → **2–3 s.f. with units**. The **scale-up** is the most-lost mark.

Titration	Ratio	Answers seen
$\text{MnO}_4^- / \text{Fe}^{2+}$	1 : 5	0.356 g $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ per tablet; 5.00% Fe^{2+} in lawn sand

Titration	Ratio	Answers seen
$\text{MnO}_4^- / \text{V}^{3+}$	2 : 5	0.202 mol dm ⁻³ (3 s.f. required)
$\text{MnO}_4^- / \text{C}_2\text{O}_4^{2-}$	2 : 5	x = 2 in $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$
$\text{Cu}^{2+} / \text{S}_2\text{O}_3^{2-}$ ($2\text{Cu}^{2+} \rightarrow \text{I}_2 \rightarrow 2\text{S}_2\text{O}_3^{2-}$)	1 : 1	46.6%, 63.3%, 65.0% Cu in brass
$\text{NaOH} / (\text{COOH})_2$	2 : 1	0.465% ethanedioic acid in rhubarb
$\text{HCl} / \text{Ca}(\text{OH})_2$	2 : 1 — $[\text{OH}^-] = 2 \times [\text{Ca}(\text{OH})_2]$	$K_c = 2.74 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$

Back titration: moles added – moles left (from the titration, **× scale-up**) = moles that reacted → ÷ **reaction ratio**.

- $(\text{NH}_4)_2\text{SO}_4 + 2\text{NaOH}$: $0.0200 - 0.006375 = 0.013625 \rightarrow \div 2 \rightarrow 15.7\%$. (Missing $\times 10 \rightarrow 22.2\%$; missing $\div 2 \rightarrow 31.3\%$.)
- NH_4X : $0.0500 - 0.0268 = 0.0232 \text{ mol} \rightarrow M = 97.8 \rightarrow X = \text{Br}$.
- KClO_3 with excess Fe^{2+} (1 : 6): $0.01125 - 5 \times 4.625 \times 10^{-4} \rightarrow \div 6 \rightarrow 0.0298 \text{ mol dm}^{-3}$.

Errors and their effect — always state the direction and the knock-on:

- NH_3 **not fully boiled off** → reacts with the HCl → **titre larger** → calculated % **lower**.
- HNO_3 **not neutralised** in the brass analysis → oxidises I^- to extra I_2 → **larger titre, higher % Cu**.
- Cl^- not removed before KMnO_4 → MnO_4^- oxidises Cl^- → larger titre → **lower** calculated KClO_3 .
- Tin left in → keeps reducing V / oxidised by MnO_4^- → **titre increases**.
- AgCl precipitate **not washed / not dry** → mass too high → moles too high.

"You must show your working" → a correct answer with no working scores **0**.

14. Enthalpy experiments

- **Polystyrene cup:** good **insulator** → less heat loss (reject "low specific heat capacity"; "prevents heat loss" ignored).
- $q = m c \Delta T$ (m = mass of **solution / water**, not + solid); $\Delta H = -q \div n \rightarrow$ **sign and kJ mol⁻¹**. E.g. LiCl: $-26.1 \text{ kJ mol}^{-1}$.
- **Calorimetry for ΔT from ΔsolH :** $\Delta T = q \div (m c)$.
- **Cooling-correction method (5 marks):** record temperature **every 30 s for ~3 min** before mixing → add at a **set time** (e.g. 3 min) → stir, record every 30 s for ~5 more minutes → plot **temperature (y) vs time (x)** → **extrapolate both best-fit lines to the time of mixing** → read ΔT .

15. Electrochemical cells

Standard cell diagram (up to 6 marks): each metal in **1.00 mol dm⁻³** solution of **its own ion** (e.g. Mg | MgSO₄(aq), Cu | CuSO₄(aq)) · **salt bridge = filter paper soaked in saturated KNO₃(aq)** · **high-resistance voltmeter** (reject "voltameter", battery, ammeter) · **298 K**. For an ion / ion couple (Fe³⁺ / Fe²⁺, V³⁺ / V²⁺) use a **platinum** electrode.

- **KOH / KCl in the salt bridge can be wrong:** OH⁻ **precipitates metal ions**; Cl⁻ reacts with Ag⁺.
- **E°cell = E°(positive, right) – E°(negative, left):** Zn/Cu +1.10 V; Mg/Cu **+2.71 V**; Cu/Fe³⁺ **+0.43 V**.
- **Running the cell for hours:** Cu²⁺ solution **paler**; Cu electrode **grows** (pink deposit); Mg electrode **smaller / pitted**; **emf falls**.
- **Non-standard E:** $E = E^\circ + (0.059/n) \log[\text{ion}]$ — e.g. [V³⁺] = 0.096 mol dm⁻³.
- **CuI precipitating** makes Cu²⁺ + I⁻ feasible even though E° is negative: removing Cu⁺ **shifts the equilibrium right**.

16. Other calculations

- **Mass from volume:** mass = volume × density (not Mr ÷ density).
 - **Gas volumes:** mol = V ÷ 24 000 cm³ at r.t.p. E.g. Ar of M in M(NO₃)₂·6H₂O = **65.3** → **zinc** (white salt). pV = nRT in Pa, m³, K.
 - **Kc from a saturated solution:** $K_c = [\text{Ca}^{2+}][\text{OH}^-]^2 = 0.0190 \times (0.0380)^2 = 2.74 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$. A **saturated** solution needs time to reach **equilibrium** (left for a day).
 - **Titre changes with temperature:** a lower titre at lower T → less dissolved → equilibrium moved left → **forward reaction endothermic**.
 - **Ka from a pH curve:** read pH at **half-equivalence** (show on the graph) → **Ka = 10^{-pH}** (e.g. 1.58 × 10⁻⁵). Add **smaller portions near the steep part** so the equivalence point isn't missed.
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PART D — KINETICS PRACTICALS

17. Methods and controls

Method	Key points
clock reaction ($\text{BrO}_3^-/\text{Br}^-$ with phenol + methyl red/orange; $\text{H}_2\text{O}_2/\text{I}^-$ or $\text{S}_2\text{O}_8^{2-}/\text{I}^-$ with thiosulfate + starch)	time t for a colour change; rate $\propto 1/t$
sampling + titration	remove samples with a pipette , quench , titrate
colorimetry	coloured species (phenolphthalein In^{2-} , I_2 , Br_2)
gas / mass loss	$\text{CaCO}_3 + \text{HCl}$: end when effervescence stops / solid disappears

Clock reactions — the explanations:

- **Thiosulfate** (or phenol) **reacts with the I_2 (or Br_2) as it forms**, so the colour appears only **after a fixed amount of reaction** (when it is used up). Reject "slows the reaction" / "reacts with I^- ".
- The amount of thiosulfate is **small**, so reactant concentrations **barely change** before the colour appears → **$1/t$ measures the initial rate**.
- **Keep the total volume constant** (add water) → **volume of a reagent $\propto$ its concentration**. A burette for small volumes (starch) keeps total volume constant.
- **Keep constant: temperature** (it changes k), total volume, volumes of the other reagents.
- Colour change: **colourless** → **blue-black** (starch–iodine); view against a **white background**.
- Reagents mixed quickly (a burette delivers too slowly); pour back and forth so they are **fully mixed**.

Other controls:

- **Other reactants in large excess** → their concentrations stay **effectively constant**, so the rate depends only on the one studied (just "in excess" is ignored). Then the order found is **only for that reactant** — not the overall order.
- **Quenching: NaHCO_3 neutralises the H^+ catalyst**; ice / cold water slows it; **excess KI** removes MnO_4^- instantly. NaOH can't be used for acid-catalysed iodination (alkali also catalyses it).
- **Ethanol as co-solvent** dissolves both the halogenoalkane and NaOH so they mix (not a catalyst).
- **Titre $\propto$ [reactant]**, so titre can be plotted instead of concentration.

18. Graphs (usually 3 marks)

1. **Axes the right way round**, labelled with **quantity / unit** (t / s, not T / s), a scale using $\geq$ **half the grid** in both directions.
2. **Points** within $\pm\frac{1}{2}$ –1 small square.
3. The **right line**: straight best-fit line / **straight line through the origin** / **smooth curve** (never point-to-point); **ignore the anomalous point** (e.g. a run at the wrong temperature).

Reading orders:

Graph	Conclusion
[X] or titre vs time straight line (constant gradient)	zero order in X
1/t (rate) vs volume / [X] straight line through origin	first order in X
conc vs time with constant successive half-lives (drawn on the graph)	first order
slow $\rightarrow$ fast $\rightarrow$ slow (sigmoid)	autocatalysis — a product (Mn^{2+}) is the catalyst; slows as reactants are used up

- **Rate at time t**: draw a **tangent** and find its gradient, **with units** (e.g. $8.2 \text{ cm}^3 \text{ h}^{-1}$).
- Half-lives in the **units of the axis** (s, not min).

19. Activation energy

- Plot **$\ln k$ (or $\ln(1/t)$, $\ln \text{rate}$) vs $1/T$** (T in K; reject $\ln(1/T)$) $\rightarrow$ best-fit line $\rightarrow$ **gradient = $-E_a/R$** $\rightarrow$ **$E_a = -\text{gradient} \times R$** .
- Give **E_a positive, in J or kJ mol^{-1} , to 2–3 s.f.** Answers: +56.5, +50.2, +57.5 kJ mol^{-1} .
- Two-point method: write $\ln k = -E_a/(RT) + c$ for both, **subtract** to remove c.
- **$\ln A$ is the y-intercept at $1/T = 0$** — not where the plotted axis happens to start.
- Why rate rises with T: more particles have **$E \geq E_a$** .

PART E — ORGANIC PREPARATION AND PURIFICATION

20. Heating under reflux

Why reflux (2 marks): heating increases the rate / lets the reaction go to completion; the **vertical condenser** condenses vapour and **returns it to the flask**, so **volatile (toxic / flammable) reactants and products are not lost** (e.g. ethanal is oxidised fully to ethanoic acid).

Diagram (3 marks):

1. **Round-bottomed or pear-shaped flask** containing the liquid, with heat.
 2. **Vertical condenser**, water in at the bottom, out at the top.
 3. **Open at the top, no gaps** — the apparatus would work.
- Classic "find the errors": **condenser sealed at the top** (pressure builds up) · **water flow reversed** (condenser doesn't fill) · **conical flask** · **no anti-bumping granules** · thermometer in the condenser.
 - **Heat source:** a **water bath or electric heater** for flammable liquids (ether) — never a Bunsen. (Water bath was rejected in one 2019 scheme but allowed since.)
 - **Anti-bumping granules: smooth boiling / prevent superheating / violent boiling** ("prevent bumping" alone ignored; reject "explosion").
 - **Stirring** is needed when reactants are **immiscible** — it gives **contact** between them.
-

21. Distillation and steam distillation

Simple / fractional distillation errors: conical flask → **pear-shaped / round flask** · thermometer in the liquid → **bulb level with the condenser inlet** (side arm) · **sealed collecting vessel** → open or vented · water flow reversed.

- **Distilling off as it forms** (aldehyde from alcohol) stops further oxidation.
- **Collect a range that brackets the boiling point** (e.g. 186–190 °C for b.p. 188 °C; a single value or a range ending at 188 is rejected).
- A fraction collected too low would contain an **impurity with a lower boiling point** (e.g. unreacted alcohol b.p. 131 °C vs ester 142 °C).
- **Boiling point on different days varies** because **atmospheric pressure** varies.

Steam distillation (limonene, lavender oil, valeric acid, iodobenzene, phenylamine):

- Used because the compound **decomposes at its boiling point**; with steam it distils **below** its boiling point.

- Diagram: **steam generator (liquid A = water, not "steam")** → **steam pipe into the liquid** in a round-bottomed flask → condenser → collector. A **vertical open tube (safety vent)** prevents pressure build-up.
 - The distillate is **cloudy / two layers** because the product is only **partly miscible** with water (an emulsion).
-

22. Separating funnel

- Draw a **funnel with a tap and stopper**; **two layers**, labelled by **density**: ester 0.88, limonene 0.851, lavender oil 0.885, ether → **upper layer**; iodobenzene 1.83 → **lower layer**.
 - **Washing with NaHCO_3 / Na_2CO_3** : **neutralises and removes the acid** (H_2SO_4 / carboxylic acid). **Stopper, invert, shake, then open the tap to release pressure** from CO_2 ($\text{H}^+ + \text{HCO}_3^- \rightarrow \text{CO}_2 + \text{H}_2\text{O}$). Confirm it's alkaline by testing a **sample removed** with red litmus / UI paper.
 - **Remove the stopper before draining** — otherwise air can't enter and the liquid won't flow (or air bubbles back and mixes the layers).
 - **Salting out**: adding **NaCl** lowers the organic product's solubility in water, helping the layers separate (it is **not** a drying agent).
 - **Washing with water** removes water-soluble impurities (HCl, NaOH, NaCl, excess acyl chloride by hydrolysis).
-

23. Drying

- **Liquids**: add **anhydrous MgSO_4 , Na_2SO_4 , CaCl_2 or CaSO_4** (silica gel allowed; **KOH pellets** for amines). **Reject CuSO_4 , CoCl_2 , conc H_2SO_4 , CaO .**
 - **Enough drying agent** when the liquid is **clear (no longer cloudy)** and the agent **no longer clumps** (stays free-flowing).
 - Then **decant / filter** and **distil** (not steam-distil).
 - **Solids**: **desiccator** or **warm (low-temperature) oven** or between filter papers — **not a hot oven** (decomposition / melting) and **not a solid drying agent** (can't be separated from a solid product).
-

24. Filtration and recrystallisation

Reduced-pressure (Büchner / suction) filtration — diagram (3 marks):

1. **Büchner funnel** with a **perforated base** and **flat filter paper** covering all the holes (labelled; reject fluted paper).
2. **Büchner flask with a side arm**, sealed with a **bung / collar**.
3. **Side arm to a vacuum pump / aspirator** (reject "pressure out").

- Advantages over gravity filtration: **faster** and gives a **drier** product (reject "removes more impurities").
- Paper must lie **flat and seal the holes**, otherwise mixture runs down the sides and solid is lost.

Recrystallisation — the ordered method (4–5 marks):

1. Dissolve in the **minimum volume of hot solvent** — so the solution is saturated and **as little product as possible stays dissolved on cooling** (maximises yield).
 2. **Filter while hot** (preheated funnel, so crystals don't form in it) — removes **insoluble impurities**.
 3. **Cool** (ice) so the product **crystallises**.
 4. **Filter under reduced pressure** — **soluble impurities stay in the filtrate**.
 5. **Wash with a small volume of ice-cold solvent** (removes soluble impurities on the surface **without dissolving the crystals**) and **dry** (desiccator / low oven).
- **Hot filtration = insoluble impurities; cold filtration = soluble impurities** — reversing them scores 0; "removes impurities" alone scores 0.
 - **Choosing a solvent:** the product is **very soluble when hot, sparingly soluble when cold**, and **doesn't react** with it.
 - Dissolve **below the melting point** of the product (phenyl benzoate m.p. 71 °C → dissolve at 60 °C, not 78 °C) so it doesn't melt.
 - Never "evaporate off the solvent".
-

25. Melting point and purity

- **Apparatus:** dry sample in a **capillary tube sealed at one end**, held **next to the thermometer bulb** in **melting-point apparatus / Thiele tube of oil / oil bath** (or an **aluminium heating block**). A water bath is useless above 100 °C.
 - Heat **quickly to ~10–20 °C below** the expected m.p., then **slowly**.
 - **Pure:** **sharp** (narrow range) and **matches the data-book value**. **Impure:** **lower** and **melts over a wider range** — both points are needed.
 - **Never boiling point;** never % yield or constant mass as a purity test.
-

26. Yield, excess and atom economy

1. **Mass of a liquid = volume × density** (e.g. $1.08 \times 4.0 = 4.32$ g).
2. Moles of each reactant; compare with the **ratio** to find the **limiting** reagent ("in excess" shown by numbers).
3. **Theoretical mass** = moles of limiting reagent × Mr(product) — calculate Mr carefully (methyl salicylate **152**, phenyl benzoate 198).

4. % yield = actual ÷ theoretical × 100 (2–3 s.f.; > 100% is wrong unless the question says why).

Yield issue	Accepted reason
< 100% (ester)	equilibrium / reversible reaction
< 100% (general)	side reactions (e.g. multiple nitration), incomplete reaction , loss on transfer (say from where), product left dissolved in the recrystallisation solvent
> 100%	product still damp (water / solvent retained) — not "hydrated salt"

Answers: 53%, 53.0%, 58.2%, 64%, 119% (damp), 0.82 g, 3.70 g, 6.85 g.

27. The preparations that are examined

Preparation	Points that recur
Esters (3-methylbutyl ethanoate, ethyl butanoate, methyl salicylate, propyl ethanoate)	conc H ₂ SO ₄ = catalyst (reject "initiates", "provides H ⁺ "); reflux; NaHCO ₃ wash + vent CO ₂ ; separate; dry; distil; yield limited by equilibrium
Nitration (methyl benzoate → methyl 3-nitrobenzoate)	conc HNO ₃ corrosive + oxidising ; add slowly in an ice–water bath (better contact than ice cubes) because it is exothermic — otherwise multiple nitration / side products (2- or 4-isomer, dinitro); recrystallise from ethanol / methanol
Aspirin (ethanoic anhydride + 2-hydroxybenzoic acid)	anhydride in excess ; crushed ice reacts with excess anhydride / precipitates aspirin ; Büchner; m.p.
Azo dyes / diazotisation	NaNO₂ + HCl (HNO ₂ made in situ — unstable); 0–10 °C in an ice bath, NaNO ₂ added dropwise , monitor with a thermometer — below 0 °C too slow, above 10 °C the diazonium ion decomposes : C ₆ H ₅ N ₂ ⁺ + H ₂ O → C ₆ H ₅ OH + N ₂ + H ⁺ ; KI added slowly (exothermic) for iodobenzene; –SO ₃ H groups make dyes water-soluble
Phenylamine (nitrobenzene + Sn/HCl)	Sn = reducing agent ; gloves: conc HCl corrosive , nitrobenzene toxic by skin absorption ; steam distil; salt out; ether extraction; KOH pellets; heat ether with a water bath (flammable)
Amides (phenylammonium chloride + ethanoic anhydride; ethanoyl chloride + amine)	use the solid salt (easier/safer to weigh; phenylamine toxic and corrosive); add water to ethanoyl chloride slowly — vigorous, releases toxic HCl

Preparation	Points that recur
Phenyl benzoate (phenol + benzoyl chloride)	gloves for phenol; fume cupboard (benzoyl chloride / HCl toxic by inhalation); benzoyl chloride in excess, so its volume need not be exact
[Cu(NH₃)₄]SO₄·H₂O	fume cupboard (NH ₃ toxic); ethanol precipitates the complex (less soluble than in water); wash with cold ethanol

PART F — HAZARDS AND SAFETY

28. Hazard + matching precaution (2 marks)

The **precaution** must match the **named hazard** of a **named substance**; goggles and lab coats are assumed and score nothing.

Hazard	Examples	Precaution
corrosive	conc HCl, conc HNO ₃ , H ₂ SO ₄ , NaOH, phenol, phenylamine, NH ₃	gloves
toxic (by inhalation)	SO ₂ , NO ₂ , NH ₃ , HCl fumes, Br ₂ , acyl chlorides, benzoyl chloride	fume cupboard
toxic by skin absorption	nitrobenzene, phenol	gloves
flammable	ether, ethanol, hexane, organic solvents	no naked flames ; water bath / electric heater
oxidising	conc HNO ₃ , NO ₂ , metal nitrates, KMnO ₄	keep away from combustible materials
carcinogenic / long-term health hazard	phenylamine, benzene	fume cupboard, gloves, minimum quantities
hazardous to the environment	organic waste, phenylamine	don't pour down the sink — collect as waste

- **Name hazards exactly:** "oxidising" (reject "oxidative", "oxidised"); "**corrosive**" for conc acids (reject "flammable"); "irritant", "harmful", "dangerous" are usually ignored.
 - Mismatches score 0: "toxic, so wear gloves" (unless toxic by skin absorption), "just fume cupboard" with no hazard.
 - **Explanations:** cold water / slow addition because the reaction is **exothermic** (reject "explosive"); generate SO₂ **in situ** so only a small, controlled amount is made; use the **solid salt** rather than a toxic liquid.
 - **Hazard symbols** to recognise: corrosive (liquid dripping onto a hand and a surface), toxic (skull and crossbones), oxidising (flame over a circle), flammable, health hazard, environment.
-

PART G — EXAM TECHNIQUE

29. Examiner rulings that decide marks

- **Formula asked** → give the formula (Cl^- , not "chloride"); **name asked** → give the name. If you give both, **both must be right**.
 - **Charges and oxidation states** are required: Cr^{3+} , $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, "iron(II) nitrate". "Metal" for "ion" is penalised.
 - **Gas vs ion**: NH_3 (gas) vs NH_4^+ (ion); HCl (fumes) vs Cl^- .
 - **Ionic equations**: no spectator ions or uncanceled electrons; state symbols only if asked (then a separate mark).
 - **Sig figs**: "ignore s.f. except 1 s.f." is typical — but obey "to 3 s.f."; **don't round early** (5.86 cm^3 scores, 5.8 doesn't); titres and mean titres to **2 d.p.**
 - **Units** on final answers (g, %, mol dm^{-3} , kJ mol^{-1} , $\text{cm}^3 \text{ h}^{-1}$); E_a **positive**; ΔH with **sign**.
 - **Dependent marks**: an observation mark usually needs the **correct reagent** (or a near miss); a precaution needs the **correct hazard**.
 - **TE** is generous through calculations — but not if the answer becomes impossible (% > 100, pH on the wrong side of 7).
-

30. The 25 most lost marks

1. Not writing "**precipitate**"; calling a solution a precipitate.
2. **Purple** for the KMnO_4 end point (it's **pale pink**).
3. "**Blue**" instead of **deep / dark blue** for copper–ammonia; $[\text{Cu}(\text{NH}_3)_6]^{2+}$.
4. "**Yellow**" instead of **pale yellow / straw** for when to add starch.
5. "**Clear**" instead of colourless; "**bright yellow**" for iodoform; **brick red** for DNPH; **lilac** for V^{2+} ; **red** for cobalt chloride paper.
6. Swapping **misty fumes** (HCl) and **white smoke** (NH_4Cl); "brown **smoke**".
7. **H_2SO_4** to acidify a sulfate test; **HCl** before AgNO_3 or in KMnO_4 titrations.
8. "**Dissolves**" without saying **in excess**.
9. **Ligand exchange** for precipitate formation (it's **deprotonation**).
10. Averaging **non-concordant** titres; mean not to 2 d.p.
11. Forgetting **$\times 10$ scale-up**, the **reaction ratio**, or $[\text{OH}^-] = 2c$ for $\text{Ca}(\text{OH})_2$.
12. Forgetting **$\times 2$** in a burette or temperature-change uncertainty.
13. **Mass = Mr ÷ density** instead of **volume $\times$ density**.

14. **Conical flask** in reflux / distillation; **sealed** apparatus; **water flow reversed**; thermometer bulb in the liquid.
15. **Fluted** filter paper or no side arm in a Büchner diagram.
16. **Hot / cold filtration** reversed, or "removes impurities" without soluble / insoluble.
17. Washing removes "**insoluble**" impurities (it removes **soluble** ones).
18. **Boiling point** as a purity test; only "lower" **or** only "range" for an impure m.p.
19. **CuSO₄ / conc H₂SO₄** as a drying agent; **hot oven** for crystals.
20. "**Prevent bumping**", "**to keep cool**", "**increase rate**", "**explosive**" — vague reasons that are ignored or rejected.
21. H₂SO₄ as "**initiator**" or just "provides H⁺" (it's a **catalyst**).
22. "**In excess**" without "so its concentration stays constant / rate depends only on...".
23. **Negative Ea**; half-lives in the wrong units; no construction lines on the graph.
24. Hazard and precaution **mismatched**; "oxidative"; generic "wear goggles".
25. **Measuring cylinder** for an accurate aliquot (or a pipette for an excess reagent).

31. Per-paper map

Series	Inorganic	Organic tests / spectra	Quantitative	Kinetics	Preparation
Jun 2019*	flame tests; Mn, Zn, Fe(II) hydroxides	C ₆ H ₁₂ O isomers; NMR	KMnO ₄ / Fe ²⁺ tablets; uncertainty	—	banana ester: reflux, NaHCO ₃ , distil
Oct 2019*	Mohr's salt; x = 6	cinnamic acid; MS, NMR	KMnO ₄ / V ³⁺ ; Zn / V cell, Nernst	—	iodobenzene: diazotisation, steam distillation
Oct 2020	copper(II) chromate(VI)	C ₄ H ₈ O ₂ ester / acids; MS, NMR	—	BrO ₃ ⁻ clock; 1/t graph	aspirin: reflux errors, Büchner, m.p.
Jan 2021	Cu ²⁺ , CO ₃ ²⁻ , BaCl ₂ ; EDTA	—	KClO ₃ : gravimetric + back titration	—	azo dye: nitration, recrystallisation
Jun 2021	vanadium colours; Cu chloro complexes; NH ₄ ⁺ , ethanoate	six C ₄ compounds; NMR; Thiele tube	combustion analysis	—	[Cu(NH ₃) ₄]SO ₄ : Büchner, 119%

Series	Inorganic	Organic tests / spectra	Quantitative	Kinetics	Preparation
Oct 2021	Cu^{2+} tests; Cu / Fe^{3+} cell	propanal; amine	brass iodometric 46.6%; ΔsolH LiCl, cooling curve	—	butyl benzoate hydrolysis; distillation errors
Jan 2022	chromium flowchart	hexane-2,5-dione; esters; NMR	—	BrO_3^- clock; graph	phenylamine: Sn/HCl, steam distil, ether
Jun 2022	CuCl_2 ; $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$	$\text{C}_5\text{H}_{10}\text{O}$ isomers; DNPH m.p.	brass 63.3%; volumetric flask	—	limonene: steam distillation, polarimeter
Oct 2022	cobalt complexes; water of crystallisation	ethyl butanoate NMR	$(\text{NH}_4)_2\text{SO}_4$ back titration 15.7%	ester hydrolysis; half-life; tangent	ethyl butanoate: separating, drying
Jan 2023	Fe^{2+} / Cr^{3+} ; halides with NH_3	—	lawn sand KMnO_4 5.00%	$(\text{CH}_3)_3\text{CCl}$ quench; half-life	amide: reflux, Büchner, m.p.
Jun 2023	Cr^{3+} / Cl^- ; FeSO_4	acyl chloride; propanoic acid; ^{13}C	—	I_2 + propanone; zero order	methyl 3-nitrobenzoate: 53.0%
Oct 2023	$\text{NH}_4\text{Cr}(\text{SO}_4)_2$	CH_3COCOOH tests	ethanedioic acid; $x = 2$	H_2O_2 / I^- clock	nitration: 58.2%
Jan 2024	Cu, Zn, Fe(III), Mn; halides; nitrate	C_3 compounds	V oxidation state by KMnO_4	—	phenylethanamide; IR
Jun 2024	$\text{Cr(III)} \rightarrow$ chromate $\rightarrow$ dichromate	hydroxy-aldehyde; NMR	NH_4X back titration $\rightarrow$ Br	MnO_4^- / ethanedioate autocatalysis	methyl salicylate: 64%
Oct 2024	BaI_2 ; $\text{NH}_4\text{Fe}(\text{SO}_4)_2$; Fe^{3+} / I^-	—	$(\text{COOH})_2 \cdot x\text{H}_2\text{O}$; K_a method	$\text{S}_2\text{O}_8^{2-}$ / I^- clock; two-point E_a	azo dye: 0–5 °C, Büchner, m.p.
Jan 2025	Cr(III), Büchner diagram	$\text{C}_8\text{H}_8\text{O}$ from combustion	—	phenolphthalein fading; Arrhenius	dye: reflux faults, recrystallisation

Series	Inorganic	Organic tests / spectra	Quantitative	Kinetics	Preparation
Jun 2025	MnCl ₂ ·4H ₂ O; Mohr's salt	—	Ca(OH) ₂ titration; Kc	iodine clock; Ea 50.2	phenyl benzoate: m.p. without electricity
Oct 2025	Zn ²⁺ ; sulfate test; Cu / Zn cell	five carbonyls / alcohols; NMR	brass 65.0%	CaCO ₃ + HCl; Ea method	lavender: steam distillation, drying
Jan 2026	Fe ²⁺ vs Cr ³⁺ ; nitrate decomposition, Ar = 65.3	distinguishing pairs; NMR	ethanoic acid pH curve, K _a	—	azo dye: 0–10 °C, recrystallisation
Jun 2026	flame tests; hydroxides (Cr, Ni, Fe, Co)	valerenic vs valeric acid	Ca(OH) ₂ stoichiometry; borax titration, ΔH	—	valerian: steam distillation, recrystallisation; nylon; Mg / Cu cell

* legacy WCH06 papers.

32. A* requirements

A* = grade A overall (≥ 480 / 600 UMS) **and** ≥ 270 / 300 across the A2 units (**WCH14 + WCH15 + WCH16**). Unit 6 repeats the same tests, techniques and calculation chains every series — it is the most learnable of the three.

Priorities: (1) cation / anion tests and TM colours (§1–§4) · (2) organic tests and distinguishing pairs (§7–§8) · (3) titration technique and calculation chains (§10–§13) · (4) recrystallisation, filtration, reflux, distillation, separating, drying, m.p. (§20–§25) · (5) kinetics practical explanations and graphs (§17–§19) · (6) hazards (§28).

33. Routine for every question

1. Observation: reagent -> what you SEE (precipitate + colour / solution colour / fumes / effervescence), before AND after.
2. Deduction: name the ion or functional group, with charge.
3. Technique: name the apparatus, say HOW (dropwise, to the mark, invert, vent) and WHY (soluble vs insoluble, exothermic, volatile).
4. Calculation: moles -> ratio -> scale-up (x10) -> answer -> units -> s.f. Mean titre from concordant titres only.
5. Graph: axes + units + half the grid; points; correct line; show construction lines (half-lives, tangents) on the graph.
6. Safety: named substance + its hazard + a matching precaution.

- **Pace:** 50 marks in 80 minutes — about 1.5 min per mark.
- Draw apparatus with a **ruler, no gaps, labelled**, and make sure it **would work** (open to air when heating, water in at the bottom).